THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 28	Perimeter = 4 × side.
2	**(a)** $5n + 2$	Write the rule using a letter-number.
3	**(d)** 16	Combine like terms first: $(1 + a)m$.
4	**(b)** $5x$	Subtract the coefficients of like terms.
5	**(b)** 14	Add inside the bracket first.
6	**(c)** $4(x + 3)$	Use brackets to keep the order of operations.
7	**(b)** 110	Fare $= f + rk$.
8	**(d)** 48	Perimeter = 4 × side.
9	**(c)** $5n + 2$	Write the rule using a letter-number.
10	**(a)** 18	Combine like terms first: $(1 + a)m$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rule: $3n + 1$ (1); For $n = 5$: $3(5) + 1 = 16$ (1).	1+1
12	Like terms: $(7 - 3)x$ and $(4 + 6)$ (1); $4x + 10$ (1).	1+1
13	Substitute: $5(6) - 3(4)$ (1); $= 18$ (1).	1+1
14	Rule: $3n + 1$ (1); For $n = 8$: $3(8) + 1 = 25$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Figure 4: $3(4) + 2 = 14$ (1); Solve $3n + 2 = 29$ (1); $n = 9$ (1).	1+1+1
16	Group the x-terms and y-terms (1); x: $5 + 3 = 8$ (1); Sum $8x + 5y$ (1).	1+1+1
17	Father $= 3s + 2$ (1); At $s = 10$: $3(10) + 2$ (1); $= 32$ years (1).	1+1+1
	OR: Figure 6: $3(6) + 2 = 20$ (1); Solve $3n + 2 = 35$ (1); $n = 11$ (1).	1+1+1
18	Group the x-terms and y-terms (1); x: $7 + 2 = 9$ (1); Sum $9x + 5y$ (1).	1+1+1
	OR: Father $= 2s + 8$ (1); At $s = 14$: $2(14) + 8$ (1); $= 36$ years (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Notebooks cost $12n$ (1); Total $= 12n + 20$ (1); At $n = 10$: 140 (1); Change $= 60$ (2).	1+1+1+2
	OR: Total $= 8r + 6$ (1); At $r = 12$: 102 (1); Solve $8r + 6 = 126$ (1); $8r = 120$ (1); $r = 15$ (1).	1+1+1+1+1
20	Notebooks cost $15n$ (1); Total $= 15n + 24$ (1); At $n = 8$: 144 (1); Change $= 56$ (2).	1+1+1+2
	OR: Total $= 10r + 4$ (1); At $r = 9$: 94 (1); Solve $10r + 4 = 144$ (1); $10r = 140$ (1); $r = 14$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $7x$	Subtract the coefficients of like terms.
2	**(d)** 21	Add inside the bracket first.
3	**(a)** $y + 2z$	Use brackets to keep the order of operations.
4	**(c)** 100	Fare $= f + rk$.
5	**(a)** 36	Perimeter = 4 × side.
6	**(b)** $2n$	Write the rule using a letter-number.
7	**(c)** 25	Combine like terms first: $(1 + a)m$.
8	**(c)** $7x$	Subtract the coefficients of like terms.
9	**(a)** 28	Add inside the bracket first.
10	**(c)** $2(a - b)$	Use brackets to keep the order of operations.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Like terms: $(9 - 5)x$ and $(2 + 8)$ (1); $4x + 10$ (1).	1+1
12	Substitute: $4(9) - 7(2)$ (1); $= 22$ (1).	1+1
13	Rule: $3n + 1$ (1); For $n = 10$: $3(10) + 1 = 31$ (1).	1+1
14	Like terms: $(12 - 7)x$ and $(5 + 3)$ (1); $5x + 8$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Figure 10: $3(10) + 2 = 32$ (1); Solve $3n + 2 = 23$ (1); $n = 7$ (1).	1+1+1
16	Group the x-terms and y-terms (1); x: $4 + 6 = 10$ (1); Sum $10x + 5y$ (1).	1+1+1
17	Father $= 3s + 5$ (1); At $s = 12$: $3(12) + 5$ (1); $= 41$ years (1).	1+1+1
	OR: Figure 8: $3(8) + 2 = 26$ (1); Solve $3n + 2 = 38$ (1); $n = 12$ (1).	1+1+1
18	Group the x-terms and y-terms (1); x: $9 + 1 = 10$ (1); Sum $10x + 3y$ (1).	1+1+1
	OR: Father $= 4s + 2$ (1); At $s = 9$: $4(9) + 2$ (1); $= 38$ years (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Notebooks cost $9n$ (1); Total $= 9n + 30$ (1); At $n = 12$: 138 (1); Change $= 62$ (2).	1+1+1+2
	OR: Total $= 6r + 8$ (1); At $r = 15$: 98 (1); Solve $6r + 8 = 128$ (1); $6r = 120$ (1); $r = 20$ (1).	1+1+1+1+1
20	Notebooks cost $20n$ (1); Total $= 20n + 24$ (1); At $n = 7$: 164 (1); Change $= 36$ (2).	1+1+1+2
	OR: Total $= 12r + 5$ (1); At $r = 10$: 125 (1); Solve $12r + 5 = 101$ (1); $12r = 96$ (1); $r = 8$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 140	Fare $= f + rk$.
2	**(a)** 60	Perimeter = 4 × side.
3	**(c)** $2n - 1$	Write the rule using a letter-number.
4	**(d)** 24	Combine like terms first: $(1 + a)m$.
5	**(d)** $9x$	Subtract the coefficients of like terms.
6	**(b)** 25	Add inside the bracket first.
7	**(a)** $pq + 5$	Product first, then add.
8	**(a)** 84	Fare $= f + rk$.
9	**(d)** 80	Perimeter = 4 × side.
10	**(c)** $(a + b)/2$	Brackets keep the sum together.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Substitute: $8(5) - 2(3)$ (1); $= 34$ (1).	1+1
12	Rule: $3n + 1$ (1); For $n = 12$: $3(12) + 1 = 37$ (1).	1+1
13	Like terms: $(10 - 4)x$ and $(9 + 1)$ (1); $6x + 10$ (1).	1+1
14	Substitute: $6(7) - 5(6)$ (1); $= 12$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Figure 5: $3(5) + 2 = 17$ (1); Solve $3n + 2 = 47$ (1); $n = 15$ (1).	1+1+1
16	Group the x-terms and y-terms (1); x: $2 + 8 = 10$ (1); Sum $10x + 3y$ (1).	1+1+1
17	Father $= 2s + 6$ (1); At $s = 16$: $2(16) + 6$ (1); $= 38$ years (1).	1+1+1
	OR: Figure 12: $3(12) + 2 = 38$ (1); Solve $3n + 2 = 20$ (1); $n = 6$ (1).	1+1+1
18	Group the x-terms and y-terms (1); x: $6 + 5 = 11$ (1); Sum $11x + 1y$ (1).	1+1+1
	OR: Father $= 3s + 1$ (1); At $s = 11$: $3(11) + 1$ (1); $= 34$ years (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Notebooks cost $18n$ (1); Total $= 18n + 20$ (1); At $n = 9$: 182 (1); Change $= 68$ (2).	1+1+1+2
	OR: Total $= 9r + 3$ (1); At $r = 11$: 102 (1); Solve $9r + 3 = 120$ (1); $9r = 117$ (1); $r = 13$ (1).	1+1+1+1+1
20	Notebooks cost $14n$ (1); Total $= 14n + 21$ (1); At $n = 15$: 231 (1); Change $= 69$ (2).	1+1+1+2
	OR: Total $= 7r + 9$ (1); At $r = 14$: 107 (1); Solve $7r + 9 = 135$ (1); $7r = 126$ (1); $r = 18$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.